Ember

The long life of a low-mass star

Stellar evolution toward an ultra-cold helium remnant

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Current result. Ember has evolved a star of $0.1M_{\odot}$ through 3.40×10^{12} yr from a static initial model at the beginning of hydrogen burning. The star remains fully convective: convection mixes material through almost its entire mass. Its hydrogen mass fraction is 0.2034 and its effective surface temperature is 3202 K. It has not exhausted hydrogen or become a helium white dwarf. The aim is to calculate that transition and the subsequent cooling to surface temperatures of 1000, 500 and 100 K.

The main comparison is with the published $0.1M_{\odot}$ model of Laughlin, Bodenheimer and Adams (1997), using their stated values and approximate curves read from their figures. The recently reconstructed F77 program is a separate, secondary comparison. Throughout this paper, $1\text{Tyr} = 10^{12}$ yr. Reported values have at most four significant figures; inferred figure values carry less precision than directly stated or calculated numbers.

1 The star and its evolution so far

The initial hydrogen mass fraction is $X_{\mathrm{H}} = 0.7$, the initial ^{3}He fraction is zero, and the metal mass fraction is $Z = 0.02$. The metals follow the relative elemental abundances of Grevesse and Sauval's GS98 mixture. Here "metals" means elements heavier than helium. The remaining initial mass is ^{4}He. The reference holds baryonic mass fixed and omits rotation, magnetism, mass loss and external heating. Its age excludes formation and pre-main-sequence contraction.

Quantity	3.30Tyr	3.40Tyr
Radius ($R_{\odot}$)	0.1470	0.1459
Photon luminosity ($L_{\odot}$)	0.001978	0.002016
Effective surface temperature (K)	3175	3202
Central temperature (MK)	7.768	8.031
Central density ($\mathrm{g\,cm^{-3}}$)	252.3	258.3
Central hydrogen mass fraction	0.2223	0.2034
Central ^{3}He mass fraction	0.001358	0.0009703
Remaining hydrogen mass ($M_{\odot}$)	0.02223	0.02034
Convective mass fraction	1	1

The model uses 512 mass points. The calculation to 3.30Tyr accepted 1822 time steps and repeated three steps at smaller sizes. Continuing the saved model to 3.40Tyr required another 38 accepted steps and no further repeats. Both parts used the same executable and physical inputs. Their common state was checked before combining the histories into 1861 distinct saved models. The decline in ^{3}He and the recent decrease in radius occur while the star is still burning hydrogen, not during white-dwarf cooling.

2 Comparison with the published LBA97 model

The main comparison is with Laughlin, Bodenheimer and Adams [1], here abbreviated LBA97. Their section 3.2 and Figure 1 describe the full hydrogen-burning evolution of a $0.1M_{\odot}$ star and its entry into white-dwarf cooling. We use numbers stated in the text and approximate values read from the published figure. The original numerical history has not been recovered.

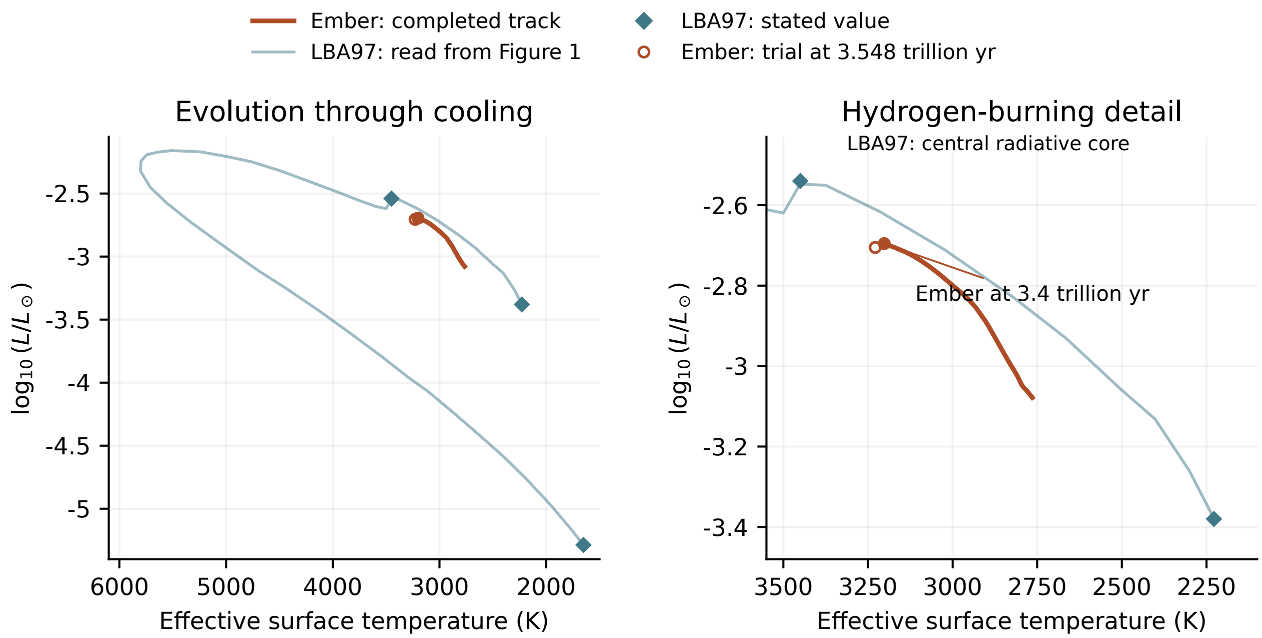

Figure 1: Ember compared with the published LBA97 evolution. The pale curve was read from Figure 1 of LBA97; diamonds mark values stated in their text. The left panel shows the full published path from the start of hydrogen burning through cooling. The right panel enlarges the region reached by Ember. The orange curve contains Ember's accepted models through 3.40Tyr; its endpoint is marked. The open orange point shows the last saved state of the refined-atmosphere trial at 3.548Tyr, which has not completed its requested interval. The original pre-main-sequence contraction and the dotted diagonal guide in LBA97 are excluded. The pale line connects 42 selected positions on the printed track; it is an approximate reconstruction, not the original numerical output. Neither temperature nor luminosity has been adjusted to make the curves agree.

Quantity	LBA97, published	Ember
Initial surface temperature (K)	2228	2764
Initial luminosity ($L_\odot$)	$\simeq 0.000417$	0.0008350
Initial central density ($\mathrm{g\,cm^{-3}}$)	309	381.5
Maximum ^{3}He mass fraction	0.0995	0.1002
Age at maximum ^{3}He (Tyr)	1.38	0.7101
Central radiative-core formation (Tyr)	5.742	Not yet established
Hydrogen mass fraction at that transition	$\simeq 0.16$	Not yet established
Highest surface temperature (K)	5807	Not yet reached

The two starting states are defined differently. LBA97 includes contraction and defines main-sequence arrival by nuclear reactions supplying all surface luminosity, after about 2 billion years. Ember starts from a static hydrogen-burning model at age zero. That clock offset is small compared with the later differences, but the initial thermal and atmospheric models also differ. Ember initially radiates about twice the published luminosity and has a hotter surface. These comparisons should not be interpreted as a measurement of the error in one isolated physical ingredient.

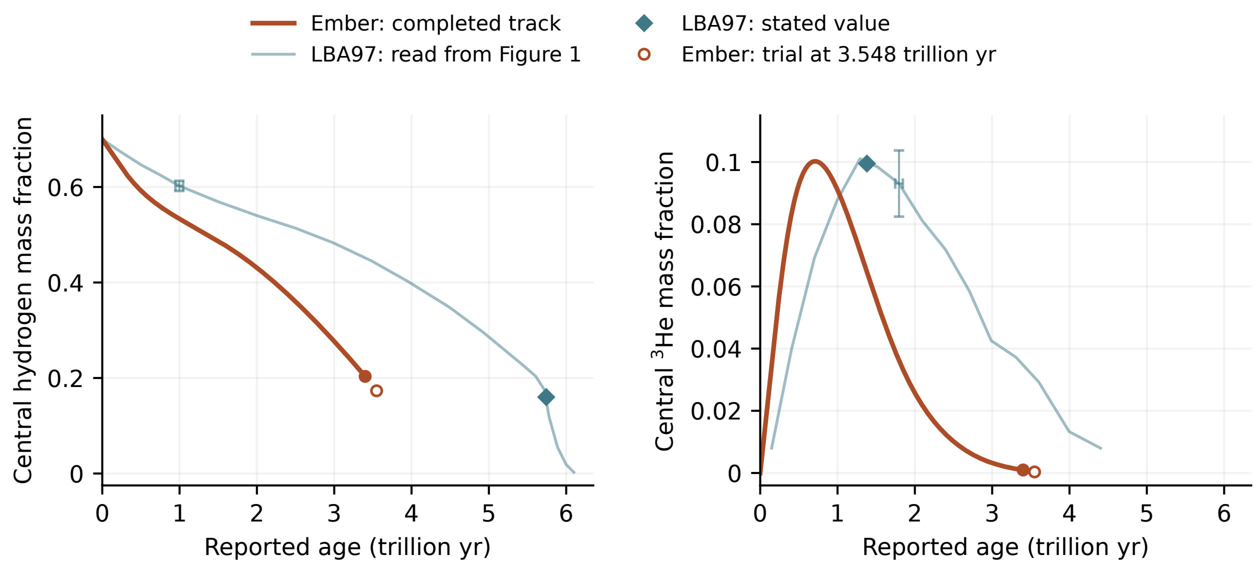

Figure 2: Hydrogen consumption and the buildup and depletion of ^{3}He. The pale curves are read from the inset of LBA97 Figure 1, with 20 hydrogen and 15 helium-3 samples. Diamonds show their stated ^{3}He maximum and the hydrogen abundance at central radiative-core formation. Orange curves are Ember's completed track; the open points show the later trial state. Ages are shown as reported, without shifting or stretching either calculation. The representative crosses correspond to four image pixels: about 0.05Tyr in age and 0.01 in mass fraction. They illustrate reading precision, not statistical or physical uncertainty. The rapid final hydrogen decline is poorly resolved in the printed inset; the helium-3 trace ends as it becomes indistinguishable from the baseline.

The maximum ^{3}He abundance agrees to within about 1%, but Ember reaches it in about half the time. Ember also consumes hydrogen faster over the interval calculated so far. Reading the published hydrogen curve gives an age of roughly 5.6Tyr at Ember's present abundance, compared with 3.40Tyr for Ember. The figure does not justify a more precise age comparison. The luminosity–temperature track alone can conceal this difference because it does not show how long the star spends at each position.

LBA97 used a grey atmosphere, approximate opacity changes as hydrogen is consumed, and older interior opacities and nuclear rates. Ember uses frequency-dependent gas atmospheres, composition-specific opacity tables, a different equation of state and updated pp-chain rates. Grain formation is still omitted from Ember's evolving atmosphere. These differences are candidates for the changed surface properties and hydrogen-consumption rate; comparisons that change one ingredient at a time are needed to separate them. Neither agreement with LBA97 nor a more expensive calculation establishes physical accuracy.

2.1 Approaching the change in convection

The completed 3.40Tyr Ember model is still fully convective. Its smallest ratio of the temperature gradient required by radiation and conduction to the adiabatic gradient is 1.336, near 21.1% of the enclosed mass; the central ratio is 3.090. For homogeneous material, a ratio below unity permits stable transport without convection. These values suggest a change first away from the center, rather than immediately forming the central radiative core described by LBA97.

A trial with the refined atmosphere table develops a stable layer between a convective center and a convective envelope. Its saved model near 3.548Tyr has 3.823% of the mass in the central convective region, 29.23% in the stable layer and 66.95% in the outer convective region. The stable shell spans approximately 19–43% of the stellar radius. Near 21.10% of the enclosed mass, the local gradient ratio has fallen to 0.9274, while the central ratio remains 1.421. Rising temperature and lower opacity favor radiation in this shell; conduction supplies only 4.144% of the local combined radiative and conductive transport capacity.

These are preliminary results. The trial stopped because a checkpoint check still capped the cumulative number of rejected time steps. Both that check and the earlier evolution-loop limit have been corrected without changing the physical accuracy tolerances. A fresh calculation and a comparison with twice as many mass points are testing the transition. Similar hydrogen abundances do not make this off-center change equivalent to LBA97's central core formation.

"Fully convective" describes where convection occurs, not the fraction of the luminosity it carries. Radiation, conduction and convection share the heat flow continuously. The model's mixing-length calculation makes the convective contribution approach zero at a stability boundary. Chemical mixing uses a stronger approximation: each connected convective region becomes uniform rapidly. Slow composition mixing is included in some marginally stable regions, but their additional heat transport still needs attention.

2.2 The reconstructed F77 program is a separate comparison

The recently reconstructed Henyey/F77 program is useful for controlled code comparisons, but it does not reproduce the LBA97 calculation. Its AJR-opacity run first flags the center as nonconvective at 3.224Tyr and $X_\mathrm{H} = 0.01756$; its Ferguson-opacity run does so at 2.946Tyr and $X_\mathrm{H} = 1.064 \times 10^{-5}$. Both differ substantially from the published 5.742Tyr and $X_\mathrm{H} \simeq 0.16$. The AJR run's highest surface temperature, 5936 K, is much closer to LBA97's 5807 K, but agreement in that one quantity does not validate the full history.

The F77 reconstruction substitutes some opacity and nuclear-rate prescriptions. At $X_\mathrm{H} = 0.2034$, it is about 76% more luminous than Ember; switching only its low-temperature opacity choice changes its luminosity by 54.5%. These comparisons establish sensitivity to the inputs, rather than identifying the source of the Ember–LBA97 difference. The separate comparison notes and supplementary F77 graphs retain these results. The F77 curves are thin and translucent, with abrupt changes drawn more faintly and all accepted data preserved.

3 Physical ingredients in Ember

Component	Treatment and present limits
Equation of state	FreeEOS 3.0 supplies pressure, internal energy and their responses to temperature, density and composition. Interpolation is based on a shared Helmholtz free energy to keep these responses consistent. The selected 64 composition tables include zero hydrogen, within a limited temperature and density range. They do not yet describe the cold, dense remnant.

Component	Treatment and present limits
Opacity	AESOPUS 2.1 provides low-temperature gas opacity and LANL TOPS provides hotter and denser conditions. The two are joined only where their available ranges permit it. Hydrogen-poor compositions have been refined and checked against additional calculations.
Hydrogen burning	A reduced ppI/ppII network follows ^{3}He out of equilibrium. It uses Solar Fusion II reaction data and finite-degeneracy Salpeter–Van Horn screening, which accounts for the plasma's effect on charged reactants. Atomic mass differences determine heat and nuclear-neutrino energy. Pep, hep, ppIII and CNO reactions are absent; metals do not react.
Heat and composition transport	Mixing-length convection uses the Ledoux criterion, including the stabilizing effect of composition gradients. Slow mixing associated with semiconvection and thermohaline instability is included. Conduction uses approximate mixtures of Ioffe/Cassisi inputs. Microscopic diffusion, settling and a separate semiconvective heat flux are absent.
Atmosphere	TLUSTY208/SYNSPEC54 calculates plane-parallel gas atmospheres in local thermodynamic equilibrium, resolving absorption and scattering over frequency. The interior joins the atmosphere at optical depth 100. The atmospheres include atomic and molecular absorption and collision-induced absorption, but no accepted condensate treatment.
Thermal neutrinos	Plasma-decay losses follow Haft, Raffelt and Weiss [4] and enter the energy equation with analytic derivatives. Other thermal-neutrino processes are not included in this choice.
Numerical evolution	Coupled implicit equations solve structure, burning and mixing. Comparing a full time step with two half steps controls time accuracy. The mass mesh is fixed. Saved models include enough internal information to continue with the same executable, tables and numerical settings.

The interior equation of state, atmosphere chemistry, opacity and conduction come from different physical sources. Their agreement therefore needs checking; accurate derivatives within one component do not prove that all components describe the same thermodynamics. FreeEOS omits potassium from the GS98 mixture, a missing mass fraction of 4.56×10^{-6}. Helium-isotope and atomic partition-function approximations also remain.

4 Why more atmosphere calculations are needed

The expensive atmosphere calculations are reusable. They are performed separately and stored as tables of boundary temperature and pressure. During stellar evolution, Ember interpolates those tables; it does not solve frequency-dependent radiative transfer at every time step. New atmosphere calculations are needed when the star reaches a composition, surface temperature or gravity not adequately covered, or when the physical treatment changes.

The completed 3.40Tyr track uses 96 atmosphere models, extending down to hydrogen fraction 0.2 and up to surface temperature 3400 K. Not every combination in the listed temperature, gravity and abundance ranges has been calculated. Ember requires the surrounding atmosphere models needed for each interpolation and its derivatives; nominal minimum and maximum values alone are insufficient.

Twelve additional atmospheres at hydrogen fraction 0.1 have converged at 3200 and 3400 K. A trial table containing all 108 atmospheres passes checks of interpolation and compatibility with the interior equation of state at those model positions. Differences between atmosphere and interior densities range from −0.01082% to +1.991%. Previously available temperature and pressure values, and their derivatives, are unchanged at 112 sampled positions. Two independent depth comparisons at the high-gravity 3200 K extremes change the matching values by less than 1.93×10^{-5} fractionally.

An atmosphere calculated independently at $X_\mathrm{H} = 0.15$, $X_3 = 0.005$, 3300 K and $\log g = 5.1$ gives a more demanding interpolation check:

Comparison with the independent atmosphere	Temperature	Gas pressure
Interpolation in all coordinates	+2.134%	−0.7868%
Composition interpolation at fixed temperature and gravity	+1.652%	−0.7842%
Temperature/gravity contribution at surrounding compositions	+0.4743%	−0.002543%

Four extra atmosphere calculations separate the composition contribution from the temperature/gravity contribution. The fractional differences combine multiplicatively, reproducing the full discrepancy. Hydrogen spacing accounts for most of the temperature error. Twelve additional atmospheres at hydrogen fraction 0.15 now form a refined table of 120 atmospheres. At the independent 0.15 atmosphere, the matching-temperature difference falls to 0.6468%; new checks at 0.125 and 0.175 differ by 0.7971% and 1.089%. The refinement also passes the original-source and interior-density checks. These remain local interpolation comparisons, not a bound on the stellar lifetime. A stellar trial using the 108-atmosphere table began losing full convection near 3.543Tyr, then stopped near 3.544Tyr after exceeding the program's allowed total number of rejected time steps. Its last accepted model had a convective mass fraction of 0.7638. The stopping rule has been changed to count consecutive rejected attempts, while retaining the cumulative count for diagnostics. No accuracy tolerance has been loosened. Fresh calculations with the refined atmosphere table and 512 and 1024 mass points are testing the transition; the figures distinguish the completed reference from the later saved trial state while those comparisons are assessed.

Interior composition tables have undergone similar checks. The first zero-hydrogen extension differed from a direct FreeEOS calculation by 2.367% in heat capacity for cool, trace-hydrogen molecular gas. Adding composition points reduced the largest pressure, energy and heat-capacity discrepancies in 528 new comparisons to 0.05213%, 0.1088%, 0.1841% and 0.1922%, respectively. Refining TOPS composition spacing reduced an initial 11.90% opacity discrepancy to a largest difference of 1.706% in the latest specified set of independent comparisons. These measure interpolation differences for the tested states, not the overall physical uncertainty in the sources.

4.1 Condensates and a hydrogen-free atmosphere

The current evolving atmosphere uses gas-only opacity. Independent chemistry checks found no condensates in the tested 3000–3200 K atmospheres, but found them in the cool upper layers of 2600–2800 K models. Effective temperature alone is therefore insufficient to decide whether grains form. The newer hydrogen-poor atmospheres still need their own checks.

Grain formation can affect radiation in two ways: it removes absorbing atoms and molecules from the gas, and any suspended grains absorb and scatter light. Earlier experiments included only the first effect, assuming the condensed material settles out. A solar-composition 2800 K comparison changed the interior matching pressure by 0.1246% and temperature by −0.02347%. These are local gas-depletion effects, not measurements of grain opacity or of a stellar lifetime change. Dusty atmosphere models become relevant around 2800 K and below [5]; grain absorption and scattering are now being added using published optical data and explicit grain-size choices. The optical calculation now passes independent scattering-program and small-particle checks. A first fixed-atmosphere experiment combines it with the computed alumina abundance. It retains absorption and scattering separately and tests several grain sizes; the atmosphere has not yet been recomputed with these opacities. Material phase, wavelength coverage and atmospheric feedback remain part of the implementation, as described in the grain-opacity notes.

Grain formation also changes thermodynamics. Earlier cool atmosphere calculations failed an enthalpy check, so the condensate treatment has not been accepted for evolution. A new calculation compares gas-only and gas-plus-grain chemical equilibria, including the redistribution of atoms among molecules and ions. At one test state the resulting heat-capacity correction is 1.175×10^5 erg g^{-1} K^{-1}, compared with 2.890×10^5 from counting grains alone. The correction still needs a consistent normalization, thermodynamic reference and treatment across phase boundaries before it can be used.

The warmer atmosphere opacity calculations currently reach about 1.010×10^4 K in their deepest layers. A trial extension to 1.500×10^4 K failed in the native helium-line broadening calculation. Reducing the bottom column is acceptable only when the atmosphere remains within the opacity range and its interior matching values no longer depend appreciably on the chosen depth.

A hydrogen-free atmosphere presents a further issue. The source programs' abundance normalization and molecular initialization currently depend on hydrogen. A direct opacity test at 4017 K shows that multiplying all number abundances by the same factor preserves composition and nearly preserves electron density, but can change individual absorption coefficients by a factor of 27.3. This one-temperature test does not measure a stellar-boundary error; it shows that a simple abundance rescaling is not yet a verified solution. A tiny nonzero hydrogen abundance cannot be labelled zero to evade the problem.

5 Energy and the cold helium remnant

The discrete energy equation used in the reference is

$$L_\gamma = L_{\mathrm{nuc,dep}} + L_{\mathrm{grav}} - L_{\nu,\mathrm{thermal}}, \qquad L_{\mathrm{nuc,rest}} = L_{\mathrm{nuc,dep}} + L_{\nu,\mathrm{nuclear}}.$$

Here L_γ is the photon luminosity, $L_{\mathrm{nuc,dep}}$ is nuclear heat left in the star, and L_{grav} is the contribution from changing internal and gravitational energy. Nuclear and thermal neutrinos are counted separately. At 3.40Tyr, in solar luminosity units,

$$L_{\mathrm{nuc,dep}} = 0.002015, \quad L_{\mathrm{grav}} = 6.548 \times 10^{-8}, \quad L_{\nu,\mathrm{nuclear}} = 4.089 \times 10^{-5}, \quad L_{\nu,\mathrm{plasma}} = 3.986 \times 10^{-11}.$$

The changing-energy term refers to the last implicit half step; the other reported diagnostics use the saved state. Before rounding, the discrete luminosity residual is below 10^{-15}. This checks the equations implemented in the code, not an independent exact energy integral or the absence of missing physical processes.

Cold degenerate electrons have a small thermal response superimposed on a large temperature-independent contribution. Subtracting the large terms had lost numerical accuracy. Paired numerical integration around the Fermi surface now retains the small response. Pure-helium tests at 100–10^4 K and densities 10^3–10^6 g cm^{-3} agree with the expected low-temperature limit to about 10^{-10} fractionally at 100 K. Electron entropy and ionic translational entropy, including mixing, are also implemented and checked against thermodynamic identities. These analytic components have not yet been combined into the production equation of state for a remnant.

An independent Ioffe EOSFI22 calculation compares liquid and solid free energies and their derivatives. Two quantum pressure-derivative errors were found and repaired in a separate copy of that source. Across sixteen forced-phase tests, the largest checked density-response discrepancy fell from 6.77 to 1.05×10^{-6} in ideal-ion pressure units after refining numerical differentiation. That establishes consistency of the tested derivatives, not which phase is stable or the physical accuracy of the formulas.

Helium quantum melting, mixture effects, latent heat and the connection to partially ionized outer layers remain unfinished. The source's default classical melting value, $\Gamma = 175$, cannot stand in for a helium phase calculation [6]. Dense, cool atmospheres also require nonideal chemistry and pressure-dependent absorption, including collision-induced absorption; refraction and dense-fluid effects need checking [7].

6 What is required to finish the calculation?

First, the atmosphere and interior inputs must cover the star's actual path toward hydrogen exhaustion. New calculations must resolve the measured interpolation discrepancies and the hydrogen-free and grain-formation issues, without extrapolating beyond available physical data.

As composition becomes stratified, conservative microscopic diffusion and settling must include the electric field that maintains charge balance and the associated energy exchange. Moving convection boundaries and thin burning layers need enough mass resolution; an adjustable mesh is required if the fixed mesh does not converge. The importance of omitted nuclear reactions, thermal-neutrino processes and semiconvective heat transport must be assessed in the conditions actually reached.

Hydrogen exhaustion and cooling require distinct measurements. We will record central hydrogen fractions crossing 10^{-3}, 10^{-4} and 10^{-5}, retaining the accepted models on either side. We will also record when nuclear heat falls below 10%, 1% and 0.1% of photon luminosity during cooling, including any later recovery. Remaining hydrogen mass and shell burning must be followed; a depleted center alone does not establish that burning has ended everywhere.

Finally, a consistent liquid/solid helium equation of state, conductive transport and a dense atmosphere must allow the star to evolve through 1000, 500 and 100 K on the descending-temperature branch. These endpoints must converge as mass resolution, time accuracy and physical tables are refined. A fitted cooling law or an extrapolated atmosphere is not the requested evolutionary calculation. Numerical error, interpolation error and uncertainty in the physical prescriptions should be reported separately.

Future mass and composition surveys will state the initial helium–metallicity relation and metal pattern explicitly. Burning hydrogen into helium at fixed metal abundance is not equivalent to forming a star at higher metallicity. External heating, encounters and hypothetical particle-physics effects belong to separately defined scenarios [2].

7 Computing time and reproducibility

The Ember calculation to 3.30Tyr used 2351 CPU seconds and 4088 seconds of elapsed time while other source calculations were running. Continuing it to 3.40Tyr used 97.00 CPU seconds. Two separate timing experiments found that reusing exactly identical nuclear calculations reduced CPU time by 12.52% and 10.84%, with identical complete output files. The combined improvement is about 22% for those experiments, not a measured whole-lifetime speedup.

The unmodified F77 AJR run used 27.37 CPU seconds through its programmed cooling stop. Ember's finer mass mesh, coupled implicit solves, time-step checks and different physics perform additional work.

Their separate contributions to the full speed difference have not been measured. An earlier atmosphere replay spent about 54% of its measured time in chemistry and only 0.0033 seconds in the dense matrix solve; that describes one atmosphere calculation, not the stellar integrator.

The GitHub repository contains code, table-generation instructions, small records of input choices and checks, this paper and the numerical data needed for its figures. The LBA97 comparison records the selected image coordinates, axis calibration and original PDF checksum; its curves can be rebuilt without the original image. The F77 figure data are extracted from accepted models and checked against the recorded output hashes. Full raw outputs, generated physical tables, saved stellar states and executable archives remain local. Instructions for recreating the figures and recovering the recorded stellar calculation accompany the paper files.

The separate build used for this report passes all 32 test suites with the locally installed physical tables. Additional atmosphere archive checks pass after accommodating two added depth descriptions while still requiring exact agreement for every previously recorded value. A fresh checkout needs the locally generated inputs for table-dependent tests and stellar calculations. Detailed input and test records retain full numerical precision and exact file identifiers so the calculations can be checked and reproduced.

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